A Morse-Theoretic Construction of the Atiyah–Hirzebruch Spectral Sequence

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Contents

Notation & Conventions

This page collects all notational conventions used in this thesis. It is to be **removed before submission**.

1. Manifolds, Maps, and the Differential

Symbol	Meaning & Convention
M, N	smooth manifolds (always assumed without boundary unless stated)
T_pM, T_p^*M	tangent / cotangent space at $p \in M$
TM, T^*M	tangent / cotangent bundle
$\mathrm{d}f$	exterior differential of $f\colon M \to \mathbb{R}$; a 1-form on M
$\mathrm{d}_pf = (\mathrm{d}f)_p$	differential of f *at the point* p; a linear map $T_pM \to \mathbb{R}$. **Rule:** write d_pf (macro `\dat{p}f`) when evaluating at a specific point. Write $\mathrm{d}f$ for the full 1-form / bundle map.
d_pF	for $F\colon M \to N$: the linear map $T_pM \to T_{F(p)}N$. Also written $(\mathrm{d}F)_p$; never write $\frac{\mathrm{d}}{\mathrm{d}x}F$ for a map between manifolds.
$\frac{\mathrm{d}}{\mathrm{d}t}$	total derivative w.r.t. a *single* real variable t (roman d, macro `\diff`). Use for curves, flows evaluated on a trajectory.
$\frac{\partial}{\partial t}$	partial derivative; use only when *at least two independent* variables/parameters are present. **Never** write $\partial/\partial t$ for the time-derivative along a flow line.
$\nabla_v f$	covariant derivative of f in direction v (Levi-Civita or chosen connection)
$\mathrm{grad}_g f$	gradient of f with respect to Riemannian metric g; characterised by $g(\mathrm{grad}_g f, -) = \mathrm{d}f$

2. Flows and Dynamical Systems

Symbol	Meaning & Convention
$\varphi_t(x)$	flow of a vector field at time t through initial point x. **Convention:** always write φ_t, not $\varphi(t,\cdot)$; think of $\varphi_t \in \mathrm{Diff}(M)$ as a one-parameter family of diffeomorphisms.
$\varphi_t = \mathrm{e}^{tX}$	flow of the vector field X; $\varphi_0 = \mathrm{id}$.
$\frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(x)$	time-derivative of the flow line $t \mapsto \varphi_t(x)$; equals $X(\varphi_t(x))$. Use roman d, never ∂.
$W^{\mathrm{u}}(p)$, $W^{\mathrm{s}}(p)$	unstable / stable manifold of a critical (rest) point p (macros `\Wu{p}`, `\Ws{p}`)

Symbol	Meaning & Convention
$\mathcal{M}(p,q)$	moduli space of (unparametrised) flow lines from p to q (macro `\Mflow{p}{q}`)

3. Restrictions and Evaluations

Core rule: distinguish carefully between a *fiber*, a *restriction to a subspace*, and a *value/evaluation.*

Symbol	Meaning & Convention
E_p	**fiber** of a vector bundle $E \to X$ over $p \in X$. **Preferred over** $E\|_{\{p\}}$ for fibers; use macro `E_p`.
$E\|_A$	**restriction** of the bundle E to the subspace $A \subseteq X$; this is itself a bundle over A. Macro: `\restr{E}{A}`.
$\xi\|_A$	section $\xi \in \Gamma(E)$ restricted to A; yields $\xi\|_A \in \Gamma(E\|_A)$.
$\xi(p)$ or ξ_p	**value** of the section ξ at the point p; an element of E_p. **Avoid** $\xi\|_p$ for this; use $\xi(p)$.
$\omega_p = \omega\|_p$	value of a differential k-form ω at p; an element of $\Lambda^k T_p^* M$. Acceptable to write $\omega\|_p$ only when the point-evaluation is being *emphasised* over the fiber.
Summary	E_p = fiber (bundle); $E\|_A$ = restricted bundle; $\xi(p)$ = value of section; ω_p = value of form.

5. Chain Complexes and Homology Theories

Three homology theories appear simultaneously; they are always decorated.

Theory	Chain complex	Boundary / Homology
Singular	$({}^{\mathcal{S}}C_*(X), {}^{\mathcal{S}}\partial)$	boundary ${}^{\mathcal{S}}\partial$; homology $H^{\mathrm{S}}H_*(X)$. Macros: `\ChS`, `\bS`, `\HS`.
CW	$({}^{\mathcal{CW}}C_*(X), {}^{\mathcal{CW}}\partial)$	boundary ${}^{\mathcal{CW}}\partial$; homology $H^{\mathrm{CW}}H_*(X)$. Macros: `\ChCW`, `\bCW`, `\HCW`.
Morse	$({}^{\mathcal{M}}C_*(f), {}^{\mathcal{M}}\partial)$	boundary ${}^{\mathcal{M}}\partial$; homology ${}^{\mathcal{M}}H_*(f)$. Macros: `\ChM`, `\bM`, `\HM`.

Connecting homomorphisms in long exact sequences: : ${}^{\mathcal{M}}\delta, {}^{\mathcal{CW}}\delta, {}^{\mathcal{S}}\delta$ with the macros: `\connM`, `\connCW`, `\connS`.
Co-homology: : For co-homology we use upper indices on the co-boundary maps: ${}^{\mathcal{M}}\mathrm{d}, {}^{\mathcal{CW}}\mathrm{d}, {}^{\mathcal{S}}\mathrm{d}$ with the macros: `\cbM`, `\cbCW`, `\cbS`. And for the co-homology we use the homlogie macros with upper indices.

Theory	Chain complex	Boundary / Homology
Filtration: $F^p\mathcal{S}C_*(X)$ for the filtration of the singular complex; boundary is still ∂.		

6. Vector Bundles and K-Theory

Symbol	Meaning
$\mathrm{Vect}(X)$	set of isomorphism classes of complex vector bundles over X (macro `\Kvect{X}`)
$[E]$	stable isomorphism class / element in $\mathrm{K}(X)$
$\mathrm{K}(X) = \mathrm{KU}(X)$	complex K-theory (default); macro `\K`, `\KU`
$\mathrm{KO}(X)$	real K-theory; macro `\KO`
$\tilde{\mathrm{K}}(X)$	reduced complex K-theory
$E \oplus F,\ E \otimes F$	Whitney sum / tensor product of bundles
$E\|_A$	restriction of bundle $E \to X$ to subspace A (same convention as §3)
E_p	fiber of E over p (same convention as §3)

7. Spectral Sequences

Symbol	Meaning
$E^r_{p,q}$	(p,q)-entry on the r-th page of a spectral sequence; macro `\Epq{r}{p,q}`
d^r	differential on the r-th page; $d^r\colon E^r_{p,q} \to E^r_{p-r,\,q+r-1}$ (cohomological convention); macro `\drpq{r}`
AHSS	Atiyah–Hirzebruch spectral sequence: $E^2_{p,q} \cong H^p(X; \mathrm{K}^q(\mathrm{pt})) \Rightarrow \mathrm{K}^{p+q}(X)$
$E^\infty_{p,q}$	abutment / E_∞-page

8. Frequently Used Macros (Quick Reference)

Macro	Output	Use
`\diff`	d	exterior deriv. / total deriv.
`\dat{p}`	d_p	deriv. at point p
`\restr{E}{A}`	$E\|_A$	restriction
`\Wu{p}`	$W^{\mathrm{u}}(p)$	unstable manifold

Macro	**Output**	**Use**
`\Ws{p}`	$W^{\mathrm{s}}(p)$	stable manifold
`\Mflow{p}{q}`	$\mathcal{M}(p,q)$	moduli of flow lines
`\ChM`	${}^{\mathcal{M}}C_*$	Morse chain complex
`\bM`	${}^{\mathcal{M}}\partial$	Morse boundary
`\HM`	${}^{\mathcal{M}}H_*$	Morse homology
`\ChCW`	${}^{\mathcal{CW}}C_*$	CW chain complex
`\bCW`	${}^{\mathcal{CW}}\partial$	CW boundary
`\HCW`	$H^{\mathrm{CW}}H_*$	CW homology
`\conn`	δ	connecting homom. (δ)
`\K`	K	K-theory
`\KO`	KO	real K-theory
`\KU`	KU	complex K-theory
`\Epq{r}{p,q}`	$E^r_{p,q}$	SS entry
`\drpq{r}`	d^r	SS differential
`\Crit`	Crit	critical point set
`\ind`	ind	Morse index
`\grad`	grad	gradient
`\euler`	e	Euler number (roman e)
`\ii`	i	imaginary unit (roman i)

1 Vector Bundles and Riemannian Geometry

Due to differential geometry and the theory of vector bundles being areas of mathematics used by pure mathematicians interested in analytical aspects like curvature, by topologists interested in invariants like Euler characteristics, physicists using the spaces as playgrounds for general relativity theories, and many others, there are numerous competing notational conventions. This thesis lies at the tripoint of differential geometry, dynamical systems, and algebraic topology. Hence, there is no a priori preferred notational convention. In this chapter, we establish the necessary background in the theory of vector bundles and Riemannian manifolds whilst not proving all the basics; we establish the notation used throughout this thesis. We establish manifolds and vector bundles in parallel, as they are interdependent on one another.

1.1 Differentiable Structures on Topological Manifolds

We begin by defining a topological manifold. Afterwards, we equip that space with with increasingly more structure and hence reduce the generality until we arrive at the playground of the thesis:

Definition 1.1.1 (Topological Manifold)**.** A second countable Hausdorff space M is called a **topological manifold** of dimension $m \in \mathbb{N}$, if it is locally homeomorphic to $\mathbb{R}^m$. To be precise, for all $p \in M$ there exists an open neighbourhood $U \subseteq M$ of p, an open set $V \subseteq \mathbb{R}^m$ and a map $\varphi : W \to V$ that is a homeomorphism. We call the map $\varphi : U \to V$ a **chart around** p **on** M, and φ^{-1} a **local coordinate system around** p **on** M.

Definition 1.1.2 (Differentiable Manifold)**.** Let M be a topological manifold of dimension m.

1. A **differentiable atlas of class** $r \in \mathbb{N} \cup \{\infty\}$ is a family of charts $\mathfrak{A} = (\varphi_i : U_i \to V_i)_{i \in I}$ such that
   a) $\bigcup_{i \in I} U_i = M$, meaning that (U_i) is an open covering of M.
   b) For every pair $(i,j) \in I^2$, the **transition function**:
   $$\begin{aligned} \varphi_{ij} : \varphi_j(U_i \cap U_j) &\to \varphi_i(U_i \cap U_j) \\ x &\mapsto \left(\varphi_i \circ \varphi_j^{-1}\right)(x) \end{aligned}$$
   is differentiable of class r.

   We call such an atlas a C^r-atlas.
2. Two C^r-atlases $\mathfrak{A}$ and $\mathfrak{B}$ are called **equivalent** if the family $\mathfrak{A} + \mathfrak{B} = (\varphi_i, \varphi_j)_{ij}$ is a C^r-atlas.

A **differentiable structure of class** r on M is an equivalence class c of C^r-atlases. For $r = \infty$, we call the pair (M, c) a smooth manifold.

Corollary 1.1.3. Every transition function φ_{ij}, $i, j \in I^2$ of a differentiable atlas $\mathfrak{A} = (\varphi_i)_{i \in I}$ is not merely a homeomorphism but also a diffeomorphism. This follows from the fact that $\varphi_{ij}^{-1} = \varphi_{ji}$ which provides smooth inverses.

Definition 1.1.4 (Smooth Functions)**.** Let (M, c) be a differentiable manifold of class r and $U \subseteq M$ open. We call a continuous function

$$f : U \to \mathbb{R}$$

differentiable of class r, if for any chart (and hence for all) $(\varphi_i)_{i \in I} = \mathfrak{A} \in c$ the compositions $f \circ \varphi_i^{-1}$ are differentiable of class r. For $r = \infty$ we define:

$$\mathcal{E}(U) = \{ f : U \to \mathbb{R} \text{ continouos} \mid f \text{ is differentiable of class } \infty \}.$$

Corollary 1.1.5. Let (M, c) be a smooth manifold of dimension m and $U \subseteq M$ be an open subset. With pointwise defined operations, the set $(\mathcal{E}(U), +, \cdot, \circ)$ becomes an $\mathbb{R}$–algebra. Furthermore, $\mathcal{E}$ becomes a sheaf of $\mathbb{R}$-algebras.

Proof. There is not really a need for a proof. However, it might help to work through the definition of a sheaf. First, $\mathcal{E}$ is a presheaf, where the restriction in the domain of a function gives the needed restriction homomorphism:

$$\begin{aligned} \operatorname{res}_V^U : \mathcal{E}(U) &\to \mathcal{E}(V) \\ f &\mapsto f|_V \,. \end{aligned}$$

The required properties of a presheaf are trivial. Furthermore, this gives a sheaf as the requirement of locality is trivial for functions and the property of gluing is also trivial for functions, since differentiability is a local property. □

Definition 1.1.6. If $p \in M$ is fix, $f \in \mathcal{E}(U)$ and $g \in \mathcal{E}(U')$ such that $p \in U \cap U'$ we say that f and g have the same **germ in** p, if there is another open neighborhood $W \subseteq U \cap U'$ of p such that $f|_W = g|_W$. This defines an equivalence relation $\sim_p$. An equivalence class s of local functions around p is called a **germ in** p. We write $s = f_p$, if $s = [f]$ with $f \in \mathcal{E}(U)$. We write

$$\mathcal{E}_p(M) = \left(\sum_{U \text{open}, p \in U} \mathcal{E}(U) \right) \Big/ \sim_p$$

for the set of germs, and call it the **stalk in** p. Where $\sum$ denotes the co-product (disjoint union) in **Top**.

Corollary 1.1.7. For a smooth manifold (M, c) the set $\mathcal{E}_p(M)$ inherits an $\mathbb{R}$–algebra structure from the $\mathcal{E}(U)$. Furthermore, it carries a natural (evaluation-)homomorphism:

$$\begin{aligned} \operatorname{eval}_p : \mathcal{E}_p(M) &\to \mathbb{R} \\ f_p &\mapsto f(p) =: f_p(p) \end{aligned}$$

The stalks are also local rings with maximal ideal $\mathfrak{m}_p = \ker(\operatorname{eval}_p)$. Hence, the pair $(M, \mathcal{E})$ gives us a locally ringed space.

Proof. We only need to verify the local ring structure of the stalks. An element $f_p \in \mathcal{E}_p(M)$ is invertible if and only if $f(p) \neq 0$, equivalently if $f_p \notin \ker(\mathrm{eval}_p)$. Thus $\mathcal{E}_p(M) \big/ \ker(\mathrm{eval}_p)$ is a field, which proves that $\ker(\mathrm{eval}_p)$ is maximal. □

Definition 1.1.8. Let (M, c) be a smooth manifold of dimension m and $p \in M$. We call an $\mathbb{R}$-linear map $\delta : \mathcal{E}_p(M) \to \mathbb{R}$ a **derivation**, if it satisfies the Leibniz rule:

$$\delta(f_p \cdot g_p) = \delta(f_p) g_p(p) + f_p(p)\delta(g_p) \quad \text{for all } f_p, g_p \in \mathcal{E}_p(M) .$$

We call $\mathrm{Der}_{\mathbb{R}}(\mathcal{E}_p(M), \mathbb{R})$ the set of derivations and give it a $\mathbb{R}$ vector space structure by pointwise operations. We define the **tangent space to** M **at** p to be the vector space

$$TM_p \coloneqq \mathrm{Der}_{\mathbb{R}}(\mathcal{E}_p(M), \mathbb{R}) .$$

Corollary 1.1.9. Let (M, c) be a smooth manifold of dimension m and $\varphi : U \to V$ be a chart around p with $x_0 = \varphi(p)$ (where $\varphi \in \mathfrak{A} \in c$). Then

$$\begin{aligned} \xi = \frac{\partial}{\partial x^j}\Big|_p : \mathcal{E}_p(M) &\to \mathbb{R} \\ f_p &\mapsto \xi(f_p) = \frac{\partial}{\partial x^j}\Big|_{x_0} (f \circ \varphi^{-1}) \end{aligned}$$

is well-defined and defines a tangent vector. Moreover, the family

$$\left(\frac{\partial}{\partial x^1}\Big|_p, \ldots, \frac{\partial}{\partial x^m}\Big|_p \right)$$

defines a basis of TM_p. Hence, the dimension of TM_p is m.

This is a good point to take a break from smooth manifolds and introduce the structure of vector bundles.

1.2 Vector Bundles

We start by defining families of vector spaces and similar to the chapter above equip them with increasingly more structure:

Definition 1.2.1 (Family of vector spaces)**.** Let X be a topological space. A **family of vector spaces over** X is a topological space E equipped with :

(i) a continuous function $\pi : E \to X$,

(ii) a finite dimensional vector space structure on each fiber $E_x \coloneqq \pi^{-1}(x)$ that is compatible with the subspace topology on $E_x \subseteq E$.

We call the map π the projection, E the total space and X the base space.